

ALQAI EmotionCam — ESP32-S3 TinyML Edge AI Camera



An end-to-end, edge-AI computer vision system built around the **ESP32-S3** microcontroller. The **ALQAI EmotionCam** project implements real-time, on-device facial emotion recognition (classifying **neutral**, **happy**, **sad**, and **surprise**) directly from camera feeds using **TensorFlow Lite Micro (TFLM)**.

The project spans the entire TinyML development lifecycle:

1. **Phase 1:** Low-level hardware diagnostic bring-up and verification.
2. **Phase 2:** Wireless dataset collection portal and model training/quantization framework.
3. **Phase 3:** On-device image preprocessing and real-time edge AI inference pipeline.

System Architecture

```
flowchart TD
    subgraph ESP32S3 [ESP32-S3 Edge Device]
        Cam[OV3660 Camera Sensor] -->|Capture VGA 640x480 JPEG| Dec[JPEG Decoder in PSRAM]
        Dec -->|Raw RGB888| Crop[Exact Center Crop 400x400]
        Crop -->|RGB888 Crop| Resize[Bilinear Resize to 128x128]
        Resize -->|128x128 RGB| Quant[INT8 Quantization: pixel - 128]
        Quant -->|Input Tensor [1,128,128,3]| TFLM[TensorFlow Lite Micro Interpreter]
        ModelSD[(SD Card /models/emotioncam_model.tflite)] -->|Load to PSRAM| TFLM
        TFLM -->|Execute Inference| Out[Output Logits]
        Out -->|Class Map| UI[OLED SSD1306 Display & Web Stream Server]
    end

    subgraph DevPC [Development PC / Model Training]
        DatasetFolder[(Collected SD Images)] -->|Transfer to PC| PythonTrain[MobileNetV2 Training Script]
        PythonTrain -->|Keras Float Model| PTQ[PTQ Quantization Script]
        PTQ -->|Quantized INT8 Model| ModelSD
    end
```

Hardware Configuration

The system is configured specifically for the **4D Systems ESP32-S3 Gen4** board (featuring **16 MB Flash** and **8 MB Octal PSRAM** in `qio_opi` mode).

Pin Mappings

Component	Signal	GPIO Pin	Description
OLED Display (SSD1306)	I2C SDA	GPIO 42	Hardware I2C for SSD1306
	I2C SCL	GPIO 41	Speed: 400 kHz, Address: 0x3C
MicroSD Card Reader	SD_MMC CMD	GPIO 38	Configured in high-speed 1-bit mode
	SD_MMC CLK	GPIO 39	
	SD_MMC D0	GPIO 40	Mount point: /sdcard
OV3660 Camera	D0 - D7	11, 9, 8, 10, 12, 18, 17, 16	8-bit parallel data interface
	XCLK / PCLK	15 / 13	System Clock (20 MHz) / Pixel Clock
	VSYNC / HREF	6 / 7	Frame and Line synchronization
	SIOD / SIOC	4 / 5	SCCB configuration interface
	PWDN / RESET	-1 / -1	Managed internally/tied

[!NOTE] All peripherals share the ESP32-S3’s high-speed **PSRAM (SPIRAM)** to accommodate large frame buffers, decoded RGB images, and TensorFlow Lite Micro’s tensor arena without exhausting internal SRAM.

Project Structure

```
esp32s3-tinyml-emotion-cam/
├── README.md           # Main project documentation
├── phaseOne/           # Hardware Diagnostics & Peripheral Testing
│   ├── src/main.cpp    # Low-level bring-up code
│   └── platformio.ini  # Phase 1 project configuration
├── phaseTwo/           # Data Ingestion & Model Development
│   ├── code/           # Python training, cleanup, and quantization scripts
│   │   ├── dataset/    # Raw captured dataset folders
│   │   ├── train_pretrained.py # MobileNetV2 V2 trainer
│   │   ├── convert_compare_ptq_v3.py # PTQ comparison & evaluator
│   │   ├── config.py   # Shared training parameters
│   │   └── requirements.txt
│   ├── src/            # Dataset collection firmware (Modular)
│   └── platformio.ini  # Phase 2 project configuration
├── phaseThree/         # Real-Time On-Device Inference
│   ├── include/        # TinyML application headers (AiConfig.h, ModelOps.h)
│   ├── src/            # TFLM inference pipeline source code
│   ├── tools/          # Helper script to parse models and generate operator resolvers
│   ├── partitions.csv  # Custom flash memory partitioning schema
│   ├── platformio.ini  # Phase 3 configuration (dual espidf/arduino framework)
│   └── sdkconfig.4d... # Tailored ESP-IDF menuconfig parameters
```



The Three Phases of Development

Phase 1: Hardware Diagnostics & Peripheral Verification

Low-level diagnostic firmware to test physical electrical connections and confirm the proper initialization of I2C, SPI/MMC, and parallel camera registers.

- **Flow:**
 1. Boot-up checks and updates status output on the SSD1306 OLED screen.
 2. Mounts the MicroSD card in 1-bit SD_MMC mode.
 3. Initializes the OV3660 camera and captures 3 test VGA frames.
 4. Saves test JPEGs to `/captures/phase6_0X.jpg` on the SD card.
 5. Performs raw file verification by checking for valid JPEG SOI (`0xFFD8`) and EOI (`0xFFD9`) markers.
 6. Displays a "Ready Face" animation on the OLED when all tests pass.

Phase 2: Wireless Dataset Collection & Model Training

Modular data ingestion firmware and model quantization pipeline.

- **Ingestion Web Portal:** Starts a Wi-Fi Access Point (`ALQAI-EmotionCam`) and hosts a local web application at `http://192.168.4.1` . Users can stream the live camera feed and capture images directly to corresponding emotion folders (`/neutral` , `/happy` , `/sad` , `/surprise`) on the SD card.
- **Dataset Verification:** Python scripts scan collected images, verify JPEG integrity, and repair corrupt files.
- **Model Training:** A custom convolutional neural network utilizing a pre-trained **MobileNetV2 base (Alpha=0.50)** with custom Dense classification layers is fine-tuned to classify 4 emotions.
- **Post-Training Quantization (PTQ):** Compares multiple quantization types (INT8 Softmax, INT8 Logits, UINT8 Logits) using representative data calibration to select the best option.

```
# Quantize and compare
python convert_compare_ptq_v3.py
```

Phase 3: On-Device TinyML Inference

The final deployment phase runs local real-time edge AI inference using TensorFlow Lite Micro.

- **TFLite Model Loader:** Loads the selected quantized model (`/models/emotioncam_model.tflite`) from the SD card directly into PSRAM.
- **Real-time Pipeline:**
 1. **Capture:** Grabs a camera frame from the OV3660.
 2. **JPEG Decode:** Decodes the JPEG frame into a raw RGB888 buffer in PSRAM.
 3. **Crop:** Extracts an exact **400x400 center crop** (focusing on the face).
 4. **Resize:** Resizes the cropped image to **128x128** using bilinear interpolation.
 5. **Quantize:** Maps floating-point pixel values to INT8 (`pixel - 128`).
 6. **Inference:** Invokes TFLite Micro to execute model inference.
 7. **Display:** Outputs the predicted emotion and execution times on the OLED screen and web interface.

Step-by-Step Execution Guide

1. Build and Flash Diagnostic Firmware (Phase 1)

Connect the board to your computer via its native USB CDC port.

```
cd phaseOne
pio run --target upload
pio device monitor
```

Ensure that OLED, SD card, and Camera diagnostics output `PASS` in the serial monitor.

2. Run Dataset Collection & Train the Model (Phase 2)

Build and flash the dataset firmware:

```
cd ../phaseTwo
pio run --target upload
```

1. Connect to the Wi-Fi AP: `ALQAI-EmotionCam` .
2. Navigate to `http://192.168.4.1` on your device.
3. Stream video and click the capture buttons to record images for each emotion class.
4. Insert the SD card into your computer, copy the images to `phaseTwo/code/dataset/` , and run training:

```
cd code
pip install -r requirements.txt
python repair_dataset_jpegs.py
python train_pretrained.py
python convert_compare_ptq_v3.py
```

5. Copy the generated `outputs/emotioncam_ptq_v3_int8_logits.tflite` to your SD card under `/models/emotioncam_model.tflite` .

3. Deploy On-Device Edge AI Inference (Phase 3)

Generate a clean, minimal TensorFlow operator resolver based on your model:

```
cd ../../phaseThree
python tools/generate_model_ops.py ../phaseTwo/code/outputs/emotioncam_ptq_v3_int8_logits.tflite include/ModelOps.h
```

This generates `include/ModelOps.h` with the exact ops used by the model, minimizing flash usage.

Now, compile and flash the inference firmware:

```
pio run --target upload
pio device monitor
```

Open `http://192.168.4.1` to view real-time predictions and camera feeds, or read the local predictions directly from the onboard OLED display!

Key Design Decisions & Optimizations

- **Dual Arduino + ESP-IDF Framework:** Arduino's ease of use is combined with ESP-IDF's low-level performance capabilities, enabling custom partitions, menuconfig overlays, and optimal PSRAM configuration.
- **TFLM Operator Optimization:** Instead of compiling all TensorFlow operators (which bloats binary sizes and exceeds flash limits), the `generate_model_ops.py` script ensures that only the exact operators used by the model are registered.
- **On-Device Image Processing:** Hardware-accelerated camera frames are preprocessed in PSRAM. Center crop (400x400), bilinear interpolation resizing, and integer mapping are performed with memory-aligned structures to achieve efficient execution times.
- **Stability Fixes:** Resolved critical issues such as FreeRTOS clock frequency mismatches (`CONFIG_FREERTOS_HZ=1000`) and entrypoint configurations (`CONFIG_AUTOSTART_ARDUINO=y`) to guarantee a stable 24/7 edge device runtime.

License & Credits

Developed by **ALQAI System** for ESP32-S3 TinyML implementations. Open-source under the MIT License.